Lab Activity 3.3: Construct form validation using Bootstrap and JavaScript

STUDENT INFORMATION

Name	Registration No	Date
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Learning Outcomes:

a) Construct client-side form validation using JavaScript.
b) Construct Bootstrap form components for improved user experience.
c) Perform form submission and validation errors gracefully.

Step 1: Setting Up the Document Structure & Styling

Create a new file named `passport_form.html`. We will start by defining the **Purple Theme** using CSS variables and styling the passport photo container to meet the **35mm x 45mm** requirement.

HTML

```
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>Lab 3.3: Passport Application Form</title>
    <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap
.min.css" rel="stylesheet">
<style>
    :root { --primary-purple: #6f42c1; --hover-purple: #59359a; --
light-purple: #f3e5f5; }
    body { background-color: var(--light-purple); padding: 40px 20px;
}
    .form-container { max-width: 800px; margin: auto; background:
white; padding: 40px; border-radius: 15px; border-top: 8px solid
var(--primary-purple); }
    .text-purple { color: var(--primary-purple); }
    .bg-purple { background-color: var(--primary-purple) !important;
}
    .btn-purple { background-color: var(--primary-purple); color:
white; }
```

```
    .btn-purple:hover { background-color: var(--hover-purple); color:
white; }
    .passport-photo { width: 3.5cm; height: 4.5cm; object-fit: cover;
border: 3px solid #f8f9fa; outline: 1px solid #dee2e6; display:
block; margin: 0 auto 10px auto; }
    .form-control:focus, .form-select:focus { border-color: var(--
primary-purple); box-shadow: 0 0 0 0.25rem rgba(111, 66, 193, 0.25);
}
    #rangeValue { background-color: var(--primary-purple); color:
white; padding: 2px 10px; border-radius: 10px; font-size: 0.9rem; }
    #successMessage { display: none; }
</style>
</head>
```

Step 2: Constructing the UI (Form Components)

Add the HTML body. This includes the **Success Alert**, **Passport Image**, **Floating Labels**, and **Input Groups**. Note the use of `novalidate` in the form tag to allow custom JavaScript handling.

HTML

```
<div class="container">
    <div class="form-container shadow-lg">

        <div class="mb-3">
            <a href="LAB2_1.html" class="text-decoration-none text-
purple small fw-bold">&larr; Back to Profile</a>
        </div>

        <h2 class="mb-4 text-center fw-bold text-purple">Passport
Application</h2>
        <hr class="mb-5">

        <div id="successMessage" class="alert alert-success alert-
dismissible fade show" role="alert">
            <strong>Success!</strong> Your application has been
validated and submitted gracefully.
            <button type="button" class="btn-close" data-bs-
dismiss="alert" aria-label="Close"></button>
        </div>

        <form class="row g-4 needs-validation" id="passportForm"
novalidate>

            <div class="col-12 text-center">
                <img src="img/profile.png" class="passport-photo
rounded shadow-sm" alt="Profile">
                <p class="text-muted small">Standard Passport Size
(35mm x 45mm)</p>
            </div>

            <div class="col-md-6">
                <div class="form-floating">
```

```
                    <input type="text" class="form-control"
id="firstName" placeholder="First Name" required>
                    <label for="firstName">First Name</label>
                    <div class="invalid-feedback">First name is
required.</div>
                </div>
            </div>

            <div class="col-md-6">
                <div class="form-floating">
                    <input type="text" class="form-control"
id="lastName" placeholder="Last Name" required>
                    <label for="lastName">Last Name</label>
                    <div class="invalid-feedback">Last name is
required.</div>
                </div>
            </div>

            <div class="col-md-12">
                <label for="userEmail" class="form-label fw-
bold">Contact Email</label>
                <div class="input-group has-validation">
                    <span class="input-group-text bg-purple text-
white">@</span>
                    <input type="email" class="form-control"
id="userEmail" placeholder="e.g. name@mail.com" required>
                    <div class="invalid-feedback">Please enter a
valid email address.</div>
                </div>
            </div>

            <div class="col-md-6">
                <label for="citizenStatus" class="form-label fw-
bold">Citizenship</label>
                <select class="form-select" id="citizenStatus"
required>
                    <option selected disabled value="">Select
status...</option>
                    <option>Citizen</option>
                    <option>Permanent Resident</option>
                </select>
                <div class="invalid-feedback">Selection
required.</div>
            </div>

            <div class="col-md-6">
                <label for="yearsResidency" class="form-label fw-
bold">
                    Residency Years: <span id="rangeValue">25</span>
                </label>
                <input type="range" class="form-range" min="1"
max="50" value="25" id="yearsResidency">
            </div>

            <div class="col-md-12">
                <label class="form-label d-block fw-
bold">Gender</label>
                <div class="form-check form-check-inline">
```

```
                    <input class="form-check-input" type="radio"
name="gender" id="male" required>
                    <label class="form-check-label"
for="male">Male</label>
                </div>
                <div class="form-check form-check-inline">
                    <input class="form-check-input" type="radio"
name="gender" id="female" required>
                    <label class="form-check-label"
for="female">Female</label>
                </div>
                <div class="invalid-feedback">Please select a
gender.</div>
            </div>

            <div class="col-12">
                <div class="form-check">
                  <input class="form-check-input" type="checkbox"
value="" id="invalidCheck" required>
                  <label class="form-check-label" for="invalidCheck">
                    I confirm that all information provided is
accurate.
                  </label>
                  <div class="invalid-feedback">You must agree before
submitting.</div>
                </div>
            </div>

            <div class="col-md-6">
                <div class="form-floating">
                    <input type="password" class="form-control"
id="password" placeholder="Password" required>
                    <label for="password">Password</label>
                    <div class="invalid-feedback">Please enter a
password.</div>
                </div>
            </div>

            <div class="col-md-6">
                <div class="form-floating">
                    <input type="password" class="form-control"
id="confirmPassword" placeholder="Confirm Password" required>
                    <label for="confirmPassword">Confirm
Password</label>
                    <div class="invalid-feedback"
id="confirmFeedback">Passwords do not match!</div>
                </div>
            </div>

            <div class="col-12 mt-5 d-flex gap-2">
                <button class="btn btn-purple w-75 py-3 fw-bold
shadow-sm" type="submit">SUBMIT APPLICATION</button>
                <button class="btn btn-outline-secondary w-25 py-3
fw-bold shadow-sm" type="reset" id="resetBtn">RESET</button>
            </div>

        </form>
    </div>
```

```
</div>
```

Step 3: JavaScript Logic (Validation & Interaction)

Implement the script to handle **Range Slider** updates, **Password Matching**, and **Graceful Error Handling**.

HTML

```
<script>
    (function () {
        'use strict'

        const form = document.getElementById('passportForm');
        const successDiv = document.getElementById('successMessage');
        const resetBtn = document.getElementById('resetBtn');
        const rangeInput = document.getElementById('yearsResidency');
        const rangeValue = document.getElementById('rangeValue');

        // Update range value display
        rangeInput.addEventListener('input', function() {
            rangeValue.textContent = this.value;
        });

        // Form Validation Logic
        form.addEventListener('submit', function (event) {
            if (!form.checkValidity()) {
                event.preventDefault();
                event.stopPropagation();
                successDiv.style.display = 'none';
            } else {
                event.preventDefault();
                successDiv.style.display = 'block';
                window.scrollTo({ top: 0, behavior: 'smooth' });
            }
            form.classList.add('was-validated');
        }, false);

        // Reset Logic
        resetBtn.addEventListener('click', function() {
            form.classList.remove('was-validated');
            successDiv.style.display = 'none';
            rangeValue.textContent = "25"; // Reset range display to
default
        });

    })();
    (function () {
    'use strict'

    const form = document.getElementById('passportForm');
    const password = document.getElementById('password');
    const confirmPassword =
document.getElementById('confirmPassword');

    // Fungsi untuk menyemak padanan kata laluan
```

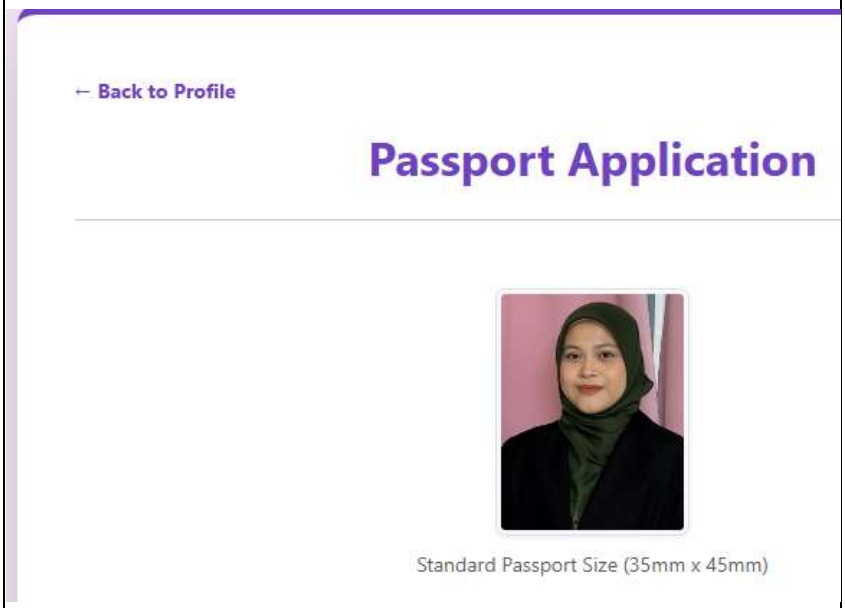
```
    function validatePassword() {
        if (password.value !== confirmPassword.value) {
            // Jika tidak sama, set ralat custom
            confirmPassword.setCustomValidity("Passwords do not
match");
        } else {
            // Jika sama, kosongkan ralat (sah)
            confirmPassword.setCustomValidity("");
        }
    }

    // Jalankan fungsi setiap kali pengguna menaip
    password.addEventListener('change', validatePassword);
    confirmPassword.addEventListener('keyup', validatePassword);

    // Kekalkan logik submit anda yang sedia ada
    form.addEventListener('submit', function (event) {
        // validatePassword dipanggil sekali lagi sebelum submit
untuk kepastian
        validatePassword();

        if (!form.checkValidity()) {
            event.preventDefault();
            event.stopPropagation();
        } else {
            event.preventDefault();
            document.getElementById('successMessage').style.display =
'block';
            window.scrollTo({ top: 0, behavior: 'smooth' });
        }
        form.classList.add('was-validated');
    }, false);

    // ... (Logik Range dan Reset anda yang lain)
})();
</script>

<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.b
undle.min.js"></script>
```

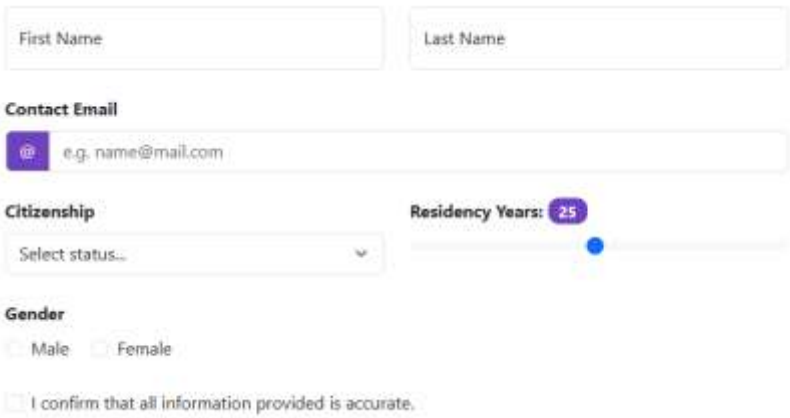
Task Recap for Students:

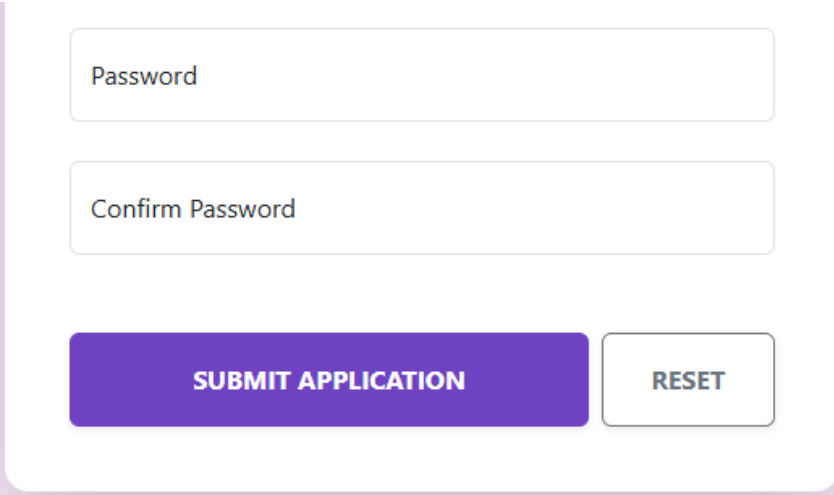
1. **Step 1:** Copy CSS to define the look and the specific **3.5cm x 4.5cm** image size.
2. **Step 2:** Construct the form using Bootstrap's grid and specialized components.
3. **Step 3:** Apply the JavaScript to enable live slider updates and ensure the passwords match before submission.
4. **Observation:** Notice how the `was-validated` class triggers red/green borders and messages gracefully without refreshing the page.

STEP-BY-STEP VALIDATION:

Please provide a **screenshot** for every completed step. For each image, write a short **observation** explaining the logic behind the code you just implemented. This helps solidify your mastery of Bootstrap!

Progress Step	**Screenshot of Output**	**Your Observation & Understanding**
Step 1		After adding the CSS and Bootstrap link in the `<head>` section, the webpage design changed. The page background became light purple and the form container appeared in the center with rounded corners and a purple border on top. **Understanding:** This step defines the styling of the webpage. The CSS variables are used to control the purple theme colors, while Bootstrap provides built-in styles that make the layout responsive and more structured.

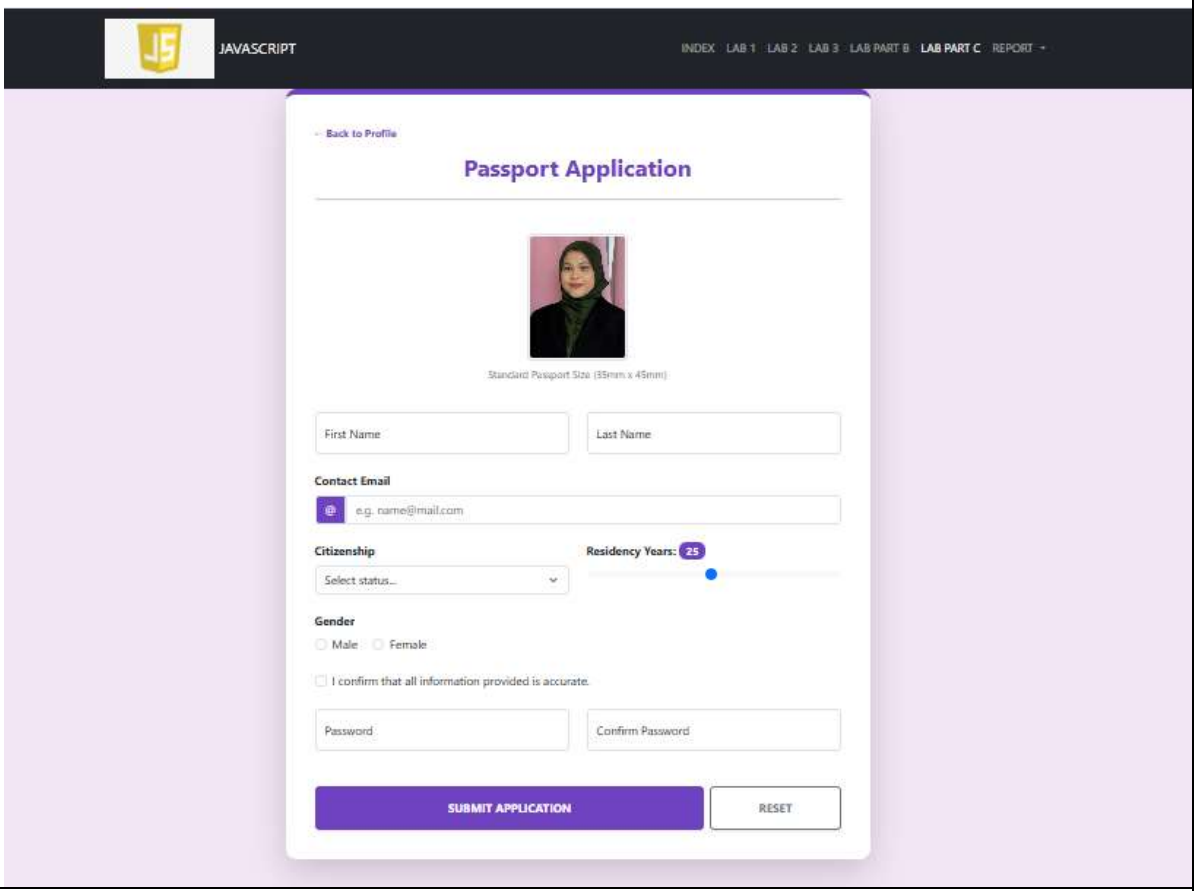
Progress Step	Screenshot of Output	Your Observation & Understanding
Step 2		After constructing the form inside the `<body>`, the Passport Application Form appeared on the page. The form includes input fields such as first name, last name, email, citizenship, gender selection, password fields, and a range slider for residency years. **Understanding:** Bootstrap components like form-floating, input groups, and grid layout help organize the form neatly. The `novalidate` attribute allows custom validation using JavaScript instead of the browser's default validation

Progress Step	Screenshot of Output	Your Observation & Understanding
Step 3		After adding the JavaScript code, the form became interactive. The residency range slider updates its value dynamically, the password confirmation checks if both passwords match, and validation messages appear when the user submits incomplete or incorrect information. **Understanding:** JavaScript handles the form validation and user interaction. The script checks if the form is valid before submission and displays a success message when all fields are filled correctly. It also prevents the page from refreshing and improves the user experience.

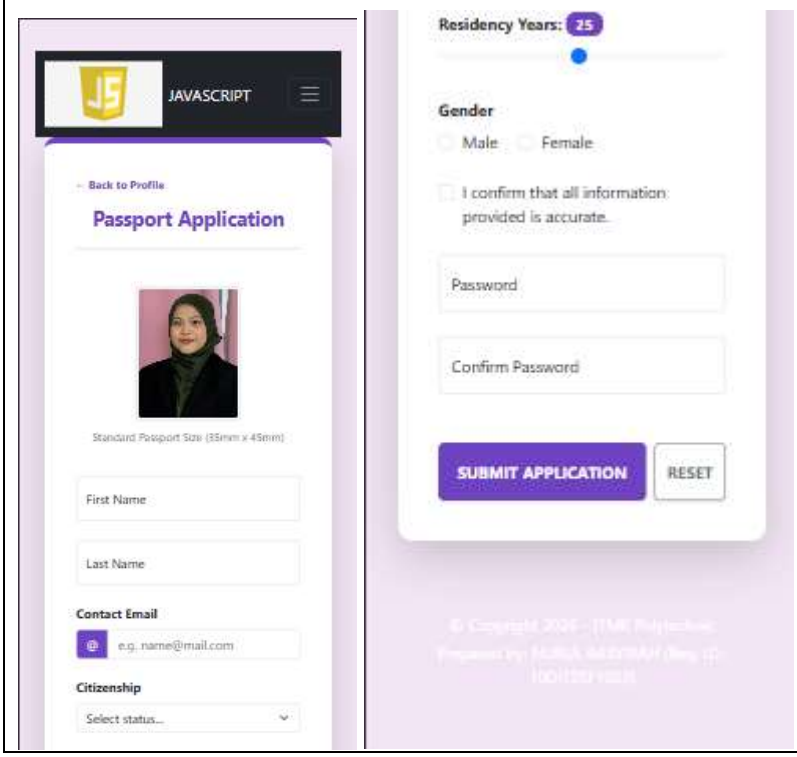
Progress Step	Screenshot of Output	Your Observation & Understanding

5. SCREENSHOT OUTPUT (COMPLETE):

Desktop View



Mobile View



6. LAB SUBMISSION TASK

Ensure you have completed all steps successfully!

1. **Compress** your folder: `RegNo_Date.zip`.
2. **Report:** Create a PDF containing all screenshots and explanations: `Report_RegNo_Date.pdf`.

Happy Coding! Success is in your hands!